

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

AXI4-Stream Ingress  
(256-bit S, K, T, r, C\_mkt, TID)

Stage 1: Ingress Registration  
& Queuing (1 Cycle)

- AXI4-Stream 256-bit Ingress: {S, K, T, r, C\_mkt, TID}
- Single-Cycle Register + TID Tag
  - No Ingress Divider (Zero Division Latency)
- Q8.24 Fixed-Point Precision

Contract

Stage 2: Analytical Seeder  
(Brenner-Subrahmanyam, 64 Cyc)

$$\sigma_0 \approx \frac{2.5066 C_{mkt}}{\sqrt{T} (S + K) / 2}$$

- Shared Digit-Recurrence  $\sqrt{T}$  (29 Cyc)
  - Forwards  $\sqrt{T}$  Directly to Stage 3
  - Non-Restoring Divider (33 Cyc)
- Obs. Liquid Error:  $|\sigma_0 - \sigma^*| / \sigma^* < 0.5\%$  (obs. < 0.08 vol in wings)

$\sigma_0, \sqrt{T}$  (Forward)

Stage 3: Core BS Datapath  
(126 Cycles, II = 1)

- Scale-Invariant Padé Ratio (S-K)/(S+K)
- 18-Cyc CORDIC Log Fallback (SRL15-Matched)
- Horner 5th-Order Poly CDF N(d1), N(d2)
- Hyperbolic CORDIC for exp(-rT) & exp(-d1<sup>2</sup>/2)
- Fully Unrolled 126-Stage Feedforward

Priority Loopback  
 $\sigma_{n+1}$  ( $|C_{BS} - C_{mkt}| > \epsilon$ )

Stage 4: NR Update & Dividers  
(33 Cycles)

- Convergence:  $|C_{BS} - C_{mkt}| \leq \$0.01$
- **u\_nr\_divider**:  $\Delta\sigma = (C_{BS} - C_{mkt}) / \nu$
- **u\_gamma\_divider**:  $\Gamma = \phi(d_1) / (S\sigma\sqrt{T})$
- Greeks:  $\Delta = N(d_1)$ ,  $\nu = S\sqrt{T}\phi(d_1)$
- TID-Scoreboard Priority Loopback

Converged IV & Greeks  
( $|C_{BS} - C_{mkt}| \leq \epsilon$ )

Stage 5: Multi-Core Arbiter  
& 128-Bit Egress (2 Cycles)

- Round-Robin 4-Core Drain Arbiter
  - 128-Bit Data Payload:
    - [127:122] Local TID (6b)
    - [121:96] Gamma Greek (26b)
    - [95:64] Vega Greek (32b)
    - [63:32] Delta Greek (32b)
    - [31:0] sigma Implied Vol (32b)
  - Sideband: Core ID (2b)
- Global Tag: {Core[1:0], TID[5:0]}

AXI4-Stream Egress  
(128b Payload + 2b Core ID Sideband)

Zero-BRAM Context Store

- 64 x 32-bit Distributed LUTRAM
- SRL32 Delay Shift Registers
  - 0 Block RAMs / UltraRAMs
- Leaves on-chip block memory for external packet & DMA buffering

Context